

SYMPLECTIC GEOMETRY

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See github.com/danimalabares/sg for the complete notes of Henrique Bursztyn's course.

1. SYMPLECTIC VECTOR SPACES

Proposition 1.1. *Every symplectic vector space (V, ω) admits a symplectic basis, i.e. a basis $e_1, \dots, e_n, f_1, \dots, f_n$ such that*

$$\omega(e_i, e_j) = 0, \quad \omega(e_i, f_j) = \delta_{ij}, \quad \omega(f_i, f_j) = 0.$$

Proof. Like Gram-Schmidt, start with any vector e_1 , then there exists another vector f_1 such that $\omega(e_1, f_1) = 1$. The space generated by e_1, f_1 is symplectic, so we get a normal decomposition... $\square$

Exercise 1.2. Let V be a vector space and ω a 2-form on V . Show that ω is nondegenerate if and only if ω^n is not zero.

Proof. If ω is nondegenerate, by Proposition 1.1 there is a symplectic basis $e_1, \dots, e_n, f_1, \dots, f_n$ such that $\omega = \sum e^i \wedge f^i$. Then

$$\omega^n = n! e^1 \wedge \dots \wedge e^n \wedge f^1 \wedge \dots \wedge f^n.$$

Hence evaluating in $e_1, \dots, e_n, f_1, \dots, f_n$ we see $\omega^n \neq 0$.

For the converse we suppose that ω is degenerate and prove that $\omega^n = 0$. It follows from the following property of interior product:

$$i_v \omega^n = n i_v \omega \wedge \omega^{n-1} \quad \forall v \in V.$$

Supposing that ω is degenerate we have $v \in V$ such that $i_v\omega = 0$, which gives $i_v\omega^n = 0$. Completing v to a basis $v_2, \dots, v_n$ we see that

$$\omega^n(v, v_2, \dots, v_n) = i_v\omega^n(v_2, \dots, v_n) = 0.$$

□

2. HAMILTONIAN VECTOR FIELDS, POISSON BRACKETS

We know that a symplectic form on a manifold M gives us a map

$$\begin{aligned}\omega^\sharp : TM &\longrightarrow T^*M \\ X &\longmapsto i_X\omega\end{aligned}$$

which says that for $f \in C^\infty(M)$,

$$i_{X_f}\omega = df.$$

Definition 2.1.

$$\begin{aligned}\{\cdot, \cdot\} : C^\infty(M) \times C^\infty(M) &\longrightarrow C^\infty(M) \\ \{f, g\} &= \omega(X_f, X_g) = df(X_g) = \mathcal{L}_{X_f}g = -\mathcal{L}_{X_g}f\end{aligned}$$

is the *Poisson bracket*.

3. SYMPLECTIC STRUCTURE OF THE COTANGENT BUNDLE

Upshot. A tangent vector to T^*M records two infinitesimal motions: the base point moves in M , and the covector moves in the cotangent fiber. The tautological 1-form ignores this second motion. It takes the covector which is the point of T^*M and evaluates it on the motion of the base point using the differential of the projection map of the bundle.

Let $\pi : T^*M \rightarrow M$ be the cotangent bundle. The *tautological 1-form* $\lambda \in \Lambda^1(T^*M)$ is defined by

$$\lambda_\xi(v) = \xi(d\pi_\xi(v)), \quad \xi \in T^*M, v \in T_\xi T^*M.$$

The canonical symplectic form on T^*M is

$$\omega_{\text{can}} = -d\lambda.$$

If (q^i) are coordinates on M and (q^i, p_i) are the induced coordinates on T^*M , then

$$\lambda = \sum_i p_i dq^i, \quad \omega_{\text{can}} = \sum_i dq^i \wedge dp_i.$$

Exercise 3.1. Let $\alpha \in \Lambda^1(M)$ and view it as a section $\alpha : M \rightarrow T^*M$.

- (1) Show that $\alpha^*\lambda = \alpha$.
- (2) Conclude that $\alpha^*\omega_{\text{can}} = -d\alpha$.
- (3) Show that the image $\alpha(M) \subset T^*M$ is Lagrangian if and only if α is closed.
- (4) In particular, show that the zero section is Lagrangian.

Proof. (1) Pick $v \in TM$.

$$\alpha^*\lambda(v) = \lambda(\alpha_*(v)) = \xi(\pi_*(\alpha_*(v)))$$

where $\alpha_*(v) \in T_\xi(M)$. Now since α is a section, $\pi \circ \alpha = \text{id}$, and thinking about it, $\xi = \alpha$!

□

Definition 3.2. Let $\varphi : M \rightarrow N$ be a diffeomorphism. The *cotangent lift* of φ is the map

$$\widehat{\varphi} : T^*M \longrightarrow T^*N$$

defined as follows. If $\xi \in T_p^*M$, then $\widehat{\varphi}(\xi)$ is the covector at $\varphi(p)$ given by

$$\widehat{\varphi}(\xi)(w) = \xi(d\varphi_p^{-1}(w)), \quad w \in T_{\varphi(p)}N.$$

In other words, the cotangent lift moves the base point by φ and transports the covector by the inverse differential.

Proposition 3.3. *The cotangent lift preserves the tautological 1-form:*

$$\widehat{\varphi}^* \lambda_N = \lambda_M.$$

In particular,

$$\widehat{\varphi}^* \omega_{\text{can}, N} = \omega_{\text{can}, M},$$

so cotangent lifts are symplectomorphisms.

Proof. Let $\xi \in T_p^*M$ and $v \in T_\xi T^*M$. The cotangent lift covers φ , so the diagram

$$\begin{array}{ccc} T^*M & \xrightarrow{\widehat{\varphi}} & T^*N \\ \pi_M \downarrow & & \downarrow \pi_N \\ M & \xrightarrow{\varphi} & N \end{array}$$

commutes. By definition of the tautological 1-form,

$$(\widehat{\varphi}^* \lambda_N)_\xi(v) = \widehat{\varphi}(\xi)(d\pi_N(d\widehat{\varphi}_\xi(v))).$$

By commutativity of the diagram,

$$d\pi_N(d\widehat{\varphi}_\xi(v)) = d\varphi_p(d\pi_M(v)).$$

Applying the definition of $\widehat{\varphi}(\xi)$, we get

$$(\widehat{\varphi}^* \lambda_N)_\xi(v) = \xi(d\pi_M(v)) = (\lambda_M)_\xi(v).$$

The statement for ω_{can} follows by applying $-d$. $\square$

This is why cotangent lifts show up so often: every diffeomorphism, or every action, on M gives a canonical symplectic diffeomorphism, or symplectic action, on T^*M . If X is a vector field on M and X_{T^*M} is the infinitesimal vector field of the lifted flow on T^*M , then

$$i_{X_{T^*M}} \omega_{\text{can}} = d(\lambda(X_{T^*M})),$$

so the cotangent lift is Hamiltonian with Hamiltonian function

$$H_X(\xi) = \xi(X_{\pi(\xi)}).$$

Exercise 3.4. Let $F : T^*M \rightarrow T^*M$ be a diffeomorphism such that

$$F^* \lambda = \lambda.$$

Show that F is the cotangent lift of a diffeomorphism $\varphi : M \rightarrow M$.

Hint: first show that F preserves the vertical distribution $\ker d\pi \subset T(T^*M)$. Then show that F sends fibers of T^*M to fibers.

4. MOSER'S TRICK

Moser's trick is an application of the following technical lemma:

Lemma 4.1. *Let M be a smooth family and X_t a one-parameter family of vector fields on M , i.e. a time-dependent vector field. Let A_t be a family of tensors indexed by the same parameter t . Then*

$$\left. \frac{d}{dt} \right|_{t=t_0} \varphi_t^* A_t = \varphi_t^* \left(\mathcal{L}_{X_t} A_t + \left. \frac{d}{dt} \right|_{t=t_0} A_t \right).$$

Theorem 4.2 (Moser's trick). *Let M be a smooth manifold and V_1, V_0 two cohomologous volume forms. Then there exists a diffeomorphism $\varphi : M \rightarrow M$ such that $\varphi^* V_1 = V_0$.*

Proof. Suppose that $V_1 - V_0 = d\eta$ for some $\eta \in \Lambda^{n-1}(M)$. To construct a time-dependent vector field consider the forms

$$V_t := tV_1 + (1-t)V_0, \quad t \in [0, 1].$$

Then we may associate a unique vector field X_t to $-\eta$ via the isomorphism $i_\bullet V_t : \mathfrak{X}(M) \rightarrow \Lambda^{n-1}(M)$. This is a time-dependent vector field, so that by Lemma 4.1 we see that the pullback forms $\varphi_t^* V_t$ are constant:

$$\begin{aligned} \left. \frac{d}{dt} \right|_{t=t_0} \varphi_t^* V_t &= \varphi_t^* \left(\mathcal{L}_{X_t} V_t + \left. \frac{d}{dt} \right|_{t=t_0} V_t \right) \\ &= \varphi_t^* \left(di_{X_t} V_t + i_{X_t} dV_t + \left. \frac{d}{dt} \right|_{t=t_0} (tV_1 + (1-t)V_0) \right) \\ &= \varphi_t^* (d(-\eta) + V_1 - V_0) \\ &= \varphi_t^* (-(V_1 - V_0) + V_1 - V_0) \\ &= 0. \end{aligned}$$

The results follows by taking $t = 1$. □

5. DARBOUX THEOREM

Theorem 5.1 (Darboux). *Let (M, ω) be a symplectic manifold and $p \in M$. There exist local coordinates $x^1, \dots, x^n, y^1, \dots, y^n$ at p such that*

$$\omega = \sum_{i=1}^n dx^i \wedge dy^i.$$

Proof. Take any chart (φ, U) at p . It's enough to show that there is a symplectomorphism $\psi : (U, \varphi^* \omega) \rightarrow (U, \omega_{\text{can}})$.

Since both $\varphi^* \omega$ and ω_{can} are closed, so is their difference, so that by Poincaré lemma, restricting the neighbourhood if necessary, we may assume that $d\eta = \varphi^* \omega - \omega_{\text{can}}$ for some 1-form η . The result follows by applying the exact same procedure as in Moser's trick as follows. Let $\omega_t := t\varphi^* \omega + (1-t)\omega_{\text{can}}$. Then ω_t is symplectic for all t . Assign to every t the vector field $X_t := \omega_t^\sharp(-\eta)$. Then apply Lemma 4.1. □

6. WEINSTEIN'S LAGRANGIAN SUBMANIFOLD THEOREM

It was Weinstein who said that everything is a Lagrangian submanifold. The following theorem says that Lagrangian submanifolds are defined intrinsically in the sense that they can all be thought of as embedded in its own cotangent bundle.

Theorem 6.1 (Weinstein's Lagrangian submanifold theorem). *If $L \subset M$ is a Lagrangian submanifold, there are neighbourhoods $L \subseteq U_0 \subset M$ and $L \subset U_1 \subset T^*L$ and a symplectomorphism $\varphi : U_0 \rightarrow U_1$.*

7. COADJOINT ORBITS AND KKS THEOREM

The coadjoint action induces orbits in $\mathfrak{g}^*$. The tangent space at any point of a coadjoint orbit is given by infinitesimal vector fields, that is,

$$T_\xi \mathcal{O} = \{u_{\mathfrak{g}^*}(\xi) : \forall u \in \mathfrak{g}\}.$$

We may introduce a symplectic form $\omega \in \Lambda^2 T_\xi^* \mathcal{O}$ on the tangent space at some point of a coadjoint orbit by

$$\omega_\xi(X, Y) := \xi([u, v]) = -u_{\mathfrak{g}^*}(\xi)v = v_{\mathfrak{g}^*}(\xi)(u)$$

which does not depend on $u, v \in \mathfrak{g}$.

Theorem 7.1 (KKS). $\xi \mapsto \omega_\xi$ is symplectic on $\mathcal{O}$.

Proof. Non degeneracy is easy. Then show that ω is G -invariant. Closedness follows from Jacobi identity. $\square$

Exercise 7.2. Let G be a Lie group. Let $X : G \rightarrow TG$ be a section of the projection $TG \rightarrow G$, not necessarily smooth. Show that if X is left invariant (i.e. $dL_g(X) = X \circ L_g$ for all $g \in G$), then X is smooth.

Proof. We have to show that $X(f)$ is a smooth function on M for any $f \in C^\infty(M)$. Indeed,

$$\begin{aligned} X(f)g &= X_g(f) \\ &= X_e \circ L_g(f) \\ &= dL_g(X_e)(f) \\ &= X_e(f \circ L_g), \end{aligned}$$

which is smooth since X_e is not varying while $f \circ L_g$ is smooth. $\square$

Exercise 7.3. For a Lie algebra $\mathfrak{g}$, there is always a canonical bilinear form $k : \mathfrak{g} \times \mathfrak{g} \rightarrow \mathbb{R}$ called the *Killing form* given by

$$k(u, v) = \text{tr}(\text{ad}_u \text{ad}_v).$$

- (1) Note that k is symmetric and check that it is Ad-invariant.
- (2) A Lie algebra is called semisimple if k is nondegenerate. Show that $\mathfrak{so}(3)$ is semisimple.

Proof. (1)

(2) Let

$$e_1 = \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, e_2 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix}, e_3 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix}$$

be a basis of $\mathfrak{so}_3$. Then

$$[e_1, e_2] = -e_3, \quad [e_1, e_3] = e_2, \quad [e_2, e_3] = -e_1.$$

To compute the trace we use Euclidean dot product identifying $\mathfrak{so}_3 = \mathbb{R}^3$.

We obtain

$$\begin{aligned} \kappa(e_1, e_2) &= \text{tr}_{\text{ad}_{e_1} \text{ad}_{e_2}} = \sum_i \langle [e_1, [e_2, e_i]], e_i \rangle \\ &= \langle [e_1, [e_2, e_1]], e_1 \rangle + \langle [e_1, [e_2, e_2]], e_2 \rangle + \langle [e_1, [e_2, e_3]], e_3 \rangle \\ &= 0, \end{aligned}$$

and similarly

$$\kappa(e_1, e_3) = 1, \quad \kappa(e_2, e_1) = 1, \quad \kappa(e_3, e_3) = -2.$$

Ok, the computations are left to the reader, but κ is should be a diagonal matrix and therefore nondegenerate. $\square$

8. HAMILTONIAN ACTIONS AND MOMENT MAP

To warm-up we define real symplectic and Hamiltonian actions.

- Definition 8.1.**
- An $\mathbb{R}$ -action is *symplectic* if $\psi_t^* \omega = \omega \iff L_X \omega = 0$, that is, X is a symplectic field.
 - An $\mathbb{R}$ -action is *hamiltonian* if X is Hamiltonian, that is, if there exists a function $H \in C^\infty(M)$ such that $X = X_H$.

To define moment map suppose there exists a map $\hat{\mu}$ such that

$$\begin{array}{ccccc} & & C^\infty(M) & & H \\ & \nearrow \hat{\mu} & \downarrow & & \downarrow \\ \mathfrak{g} & & \mathfrak{X}(M) & & X_H \\ & \searrow u & & \swarrow & \\ & & u_M = X_u & & \end{array}$$

Note that if $\hat{\mu}$ exists, it's not uniquely determined, and that we may suppose that it is linear.

It's equivalent to define $\hat{\mu}$ as a map $\hat{\mu} : \mathfrak{g} \rightarrow C^\infty(M)$ or as a map

$$\begin{aligned} \mu : M &\longrightarrow \mathfrak{g}^* \\ \hat{\mu}(u)(x) &= \langle \mu(x), u \rangle. \end{aligned}$$

- Definition 8.2.**
- A G -action is *symplectic* if $\psi_g^* \omega = \omega$ for all $g \in G$, that is, if

$$\begin{array}{ccc} \psi : G & \longrightarrow & \text{Dif}(M) \\ & \searrow & \uparrow \\ & & \text{Symp}(M, \omega) \end{array}$$

- A symplectic action of G on (M, ω) via ψ is *weakly Hamiltonian* if there exists a map $\mu : M \rightarrow \mathfrak{g}^*$ (or equivalently, $\hat{\mu} : \mathfrak{g} \rightarrow C^\infty(M)$ linear) such that

$$i_{u_M} \omega = d \langle \mu(p), u \rangle, \quad (\text{resp. } u_M = X_{\hat{\mu}(u)}).$$

- A weakly Hamiltonian action is *Hamiltonian* if it is also μ -equivariant for the action of G on M and the coadjoint action on $\mathfrak{g}^*$, that is,

$$\mu \circ \psi_g = (\text{Ad}^*)_g(\mu).$$

In this case we call μ a *moment map*.

Exercise 8.3. Let $V = \mathbb{R}^2$ with symplectic form $\omega = dx \wedge dy$. Consider the action of S^1 on V given by rotations,

$$e^{it}(x, y) = (x \cos t - y \sin t, x \sin t + y \cos t).$$

Identifying $\mathfrak{s}^1 = \mathbb{R}$, let $u = 1$.

- (1) Compute the infinitesimal vector field

$$u_V(x, y) = \left. \frac{d}{dt} \right|_{t=0} e^{it}(x, y).$$

- (2) Compute the 1-form $i_{u_V}\omega$.
- (3) Find a function $H : \mathbb{R}^2 \rightarrow \mathbb{R}$ such that

$$dH = i_{u_V}\omega.$$

- (4) Check that

$$H(v) = \frac{1}{2}\omega(uv, v).$$

Here we identify $T_v V$ with V , and uv denotes the infinitesimal action of u on v .

Proof. We have

$$\begin{aligned} u_V(x, y) &= \left. \frac{d}{dt} \right|_{t=0} (x \cos t - y \sin t, x \sin t + y \cos t) \\ &= (-y, x). \end{aligned}$$

Let $w = (a, b)$. Since

$$\omega((p, q), (a, b)) = pb - qa,$$

we get

$$\begin{aligned} (i_{u_V}\omega)(w) &= \omega((-y, x), (a, b)) \\ &= -yb - xa. \end{aligned}$$

Thus

$$i_{u_V}\omega = -x dx - y dy.$$

Therefore

$$H(x, y) = -\frac{1}{2}(x^2 + y^2)$$

satisfies

$$dH = i_{u_V}\omega.$$

Finally, we identify $T_v V$ with V . Since

$$uv = (-y, x), \quad v = (x, y),$$

we get

$$\begin{aligned} \frac{1}{2}\omega(uv, v) &= \frac{1}{2}\omega((-y, x), (x, y)) \\ &= -\frac{1}{2}(x^2 + y^2). \end{aligned}$$

Hence

$$H(v) = \frac{1}{2}\omega(uv, v).$$

□

The following exercise shows how the moment map, i.e. $\hat{\mu} = H$ looks for any linear symplectic action.

Exercise 8.4. Let (V, ω) be a symplectic vector space and let G act linearly on V preserving ω . Write

$$u_M(v) = X_u(v) = uv = \left. \frac{d}{dt} \right|_{t=0} \exp(tu)v$$

for the infinitesimal action of u on v . We identify $T_v V$ with V and, for $u \in \mathfrak{g}$, define

$$H_u(v) = \frac{1}{2}\omega(uv, v).$$

Show that

$$dH_u = i_{u_V} \omega,$$

where $u_V(v) = uv$ is the infinitesimal vector field associated to u .

Proof. Let $w \in V$. Since V is a vector space, we identify $T_v V$ with V . Thus w is a tangent vector at v , and $dH_u(v)(w)$ is the directional derivative of H_u at v in the direction w . Therefore

$$\begin{aligned} dH_u(v)(w) &= \left. \frac{d}{dt} \right|_{t=0} H_u(v + tw) \\ &= \frac{1}{2} \left. \frac{d}{dt} \right|_{t=0} \omega(u(v + tw), v + tw) \\ &= \frac{1}{2}\omega(uw, v) + \frac{1}{2}\omega(uv, w). \end{aligned}$$

Here we used that ω is bilinear: differentiating a bilinear expression means differentiating in the first slot and in the second slot.

Since the action preserves ω , we have

$$\omega(e^{tu}a, e^{tu}b) = \omega(a, b)$$

for all $a, b \in V$. Differentiating at $t = 0$ gives

$$\omega(ua, b) + \omega(a, ub) = 0.$$

Taking $a = w$ and $b = v$, we obtain

$$\omega(uw, v) + \omega(w, uv) = 0.$$

Since ω is skew-symmetric, this gives

$$\omega(uw, v) = \omega(uv, w).$$

Therefore

$$dH_u(v)(w) = \omega(uv, w).$$

But $u_V(v) = uv$, so

$$dH_u(v)(w) = (i_{u_V} \omega)_v(w).$$

Thus

$$dH_u = i_{u_V} \omega.$$

□

Exercise 8.5. Let (M, ω) be a symplectic manifold and let G act on M with moment map $\mu : M \rightarrow \mathfrak{g}^*$. Assume that 0 is a regular value of μ . Show that, for every $p \in \mu^{-1}(0)$, we have

$$T_p \mu^{-1}(0) = (T_p(Gp))^\omega.$$

Here $(T_p(Gp))^\omega$ denotes the symplectic orthogonal of the tangent space to the orbit.

Proof. For the inclusion $\subseteq$ we need to show that $\omega(v, w) = 0$ for any $v \in T_p \mu^{-1}(0)$ and $w \in T_p(G \cdot p)$. A w looks like: take a 1-parameter subgroup of G , which of course we want to see as something of the kind $\exp(tu) \cdot p$ and differentiate at $t = 0$. And of course this is the infinitesimal generator $u_M = X_u$ for $u \in \mathfrak{g}$. On the other hand, a v looks like: take a curve $\gamma \subset \mu^{-1}(0)$ and differentiate at $t = 0$.

By the moment map property we see that

$$\omega(u_M, v) = d\langle \mu, u \rangle_p(v).$$

How to compute that differential? I remember that the differential of a function can be computed by using a curve that realises the vector, i.e. $df(v) = \left. \frac{d}{dt} \right|_{t=0} f \circ \gamma(t)$. That is,

$$d\langle \mu(p), u \rangle_p v = \left. \frac{d}{dt} \right|_{t=0} \langle \mu(\gamma(t)), u \rangle = 0.$$

For the $\supseteq$ inclusion suppose that v is such that $\omega(u_M, v) = 0$ for all $u \in \mathfrak{g}$. To show that $v \in T_p \mu^{-1}(0)$ we need to show that $v \in \text{Ker } d\mu$ by the level set theorem. This is what the moment map equation says: $0 = i_{u_M} \omega(v) = d\langle \mu, u \rangle_p(v)$, and the latter differential is in fact $d(\text{ev}_u \circ \mu)$. Since ev_u is linear, its differential is again ev_u . Hence

$$d\langle \mu, u \rangle_p(v) = \langle d\mu_p(v), u \rangle.$$

Since this vanishes for every $u \in \mathfrak{g}$, the covector $d\mu_p(v) \in \mathfrak{g}^*$ vanishes identically. Thus $d\mu_p(v) = 0$. $\square$

9. NOETHER'S PRINCIPLE

Theorem 9.1 (Noether's principle). *Let (M, ω, μ) be a Hamiltonian G -space with Hamiltonian $H \in C^\infty(M)$. H is G -invariant if and only if μ is preserved by the Hamiltonian flow.*

Proof. one line $\square$

10. SYMPLECTIC QUOTIENT

In symplectic geometry we cannot always take quotients. We need to use level-sets of the moment map.

Theorem 10.1 (Marsden-Weinstein, Megre). *Let G act on (M, ω) in a Hamiltonian way with moment map $\mu : M \rightarrow \mathfrak{g}^*$. Suppose $0 \in \mathfrak{g}^*$ is a regular value. By the equivariance of μ , we have an action of G on $\mu^{-1}(0)$ given as follows. For $x \in \mu^{-1}(0)$, $\mu(\psi_g(x)) = \text{Ad}_g^*(\mu(x)) = 0$. Suppose that this action is regular (this will be defined later), that $\mu^{-1}(0)/G$ is smooth, and that the projection is a submersion. Then there exists a unique symplectic form ω_{red} in M_0 such that $i^* \omega = \pi_0^* \omega_{\text{red}}$.*

That is, the pullback of the symplectic form to the 0 level-set of μ passes to the quotient:

$$\begin{array}{ccc} G \text{ acts on } \mu^{-1}(0) & \hookrightarrow & M \\ \downarrow \pi_0 & & \\ (M_0 := \mu^{-1}(0)/G, \omega_{\text{red}}) & & \end{array}$$

A slightly more general setting is the following. Hamiltonian action of G on (M, ω) , and moment map $\mu : M \rightarrow \mathfrak{g}^*$. $\xi \in \mathfrak{g}^*$ regular, $G_\xi \subseteq G$ stabilizer of ξ . Then we have

$$\begin{array}{ccc} G_\xi \text{ acts on } \mu^{-1}(\xi) & \xrightarrow{i_\xi} & M \\ \downarrow \pi_\xi & & \\ (M_\xi = \mu^{-1}(\xi)/G_\xi, \omega_{\text{red}}) & & \end{array}$$

Exercise 10.2. Consider the group $\text{SO}(3)$ acting on $T^*\mathbb{R}^3$ by the cotangent lift of the usual action of $\text{SO}(3)$ on $\mathbb{R}^3$.

- (1) For $u \in \mathfrak{so}(3)$, compute the corresponding infinitesimal generator $u_{T^*\mathbb{R}^3} \in \mathfrak{X}(T^*\mathbb{R}^3)$.
- (2) Identify $\mathfrak{so}(3)$ with $\mathbb{R}^3$ as in Lista 5. Show that, with this identification, we have $u_{T^*\mathbb{R}^3}(q, p) = (u \times q, u \times p)$.
- (3) Identifying $\mathfrak{so}(3)^* \cong (\mathbb{R}^3)^* \cong \mathbb{R}^3$ using the usual inner product, show that the moment map for the action of $\text{SO}(3)$ on $T^*\mathbb{R}^3$ is $\mu(q, p) = q \times p$. Conclude (by Noether's theorem) that if $V \in C^\infty(\mathbb{R}^3)$ is $\text{SO}(3)$ -invariant, then the flow of the Hamiltonian $H(q, p) = \frac{p^2}{2m} + V(q)$ preserves “angular momentum” $q \times p$.

Proof. (1) To survive without getting too involved in the definitions, I observe the following. $\text{SO}(3)$ acts by rotations on $\mathbb{R}^3$. This means that the tangent vector to any 1-parameter orbit is a tangent vector to a sphere containing the 1-parameter orbit. In other words, we have an expression for the 1-parameter orbit $p \cos t + u_{\mathbb{R}^3} \sin t$ where $u_{\mathbb{R}^3}$ is the $\mathbb{R}^3$ component of $u_{T^*\mathbb{R}^3}$ at $p \in \mathbb{R}^3$. This just says that $u_{\mathbb{R}^3}$ is orthogonal to p . Since the induced map on the cotangent space is essentially given by the pushforward of the action, which is linear, then it's exactly the same — thus the $T^*\mathbb{R}^3$ component of $u_{T^*\mathbb{R}^3}$ must also be orthogonal to the covector where it is based.

Now let's get too involved in the definitions. By definition, for $(p, q) \in T^*\mathbb{R}^3$ and $u \in \mathfrak{so}_3$,

$$u_{T^*\mathbb{R}^3} = \left. \frac{d}{dt} \right|_{t=0} e^{tu}(p, q).$$

To find the integral curve $\varphi(t) = e^{tu}$ recall how it is constructed: the vector $u \in T_e G$ gives the left-invariant vector field $X(g) = d_e L_g(u)$, which has an integral curve at the identity $\varphi_u(t)$ defined by the properties $\varphi'_u(0) = u$ and $\varphi_u(0) = e$. We denote $\varphi_u(t) = e^{tu}$. Thus differentiating the integral curve by definition gives the left-invariant vector field X along φ_u , which is the differential of a linear map and thus it acts by itself, that is $X(\varphi(t)) = d_e L_{\varphi_u(t)}(u) = \varphi_u(t)u$.

Differentiating and evaluating at $t = 0$ we see that $u_{T^*\mathbb{R}^3} = (up, uq)$ since, in fact, the action on the cotangent part is also given by simple matrix multiplication.

More explicitly, let $A \in \text{SO}(3)$. The cotangent lift is encoded by

$$\begin{array}{ccc} T^*\mathbb{R}^3 & \xrightarrow{\hat{A}} & T^*\mathbb{R}^3 \\ \pi \downarrow & & \downarrow \pi \\ \mathbb{R}^3 & \xrightarrow{A} & \mathbb{R}^3 \end{array}$$

and sends (q, ξ) to $(Aq, A \cdot \xi)$, where

$$(A \cdot \xi)(v) = \xi(A^{-1}v).$$

Use the Euclidean inner product to identify covectors with vectors. Thus write

$$\xi(v) = \langle p, v \rangle.$$

Here p is the vector representing ξ by the Riesz representation theorem. By the definition of adjoint from Definition ??, we have

$$\langle A^{-1}v, p \rangle = \langle v, (A^{-1})^\dagger p \rangle.$$

Therefore

$$\begin{aligned} (A \cdot \xi)(v) &= \xi(A^{-1}v) \\ &= \langle p, A^{-1}v \rangle \\ &= \langle A^{-1}v, p \rangle \\ &= \langle v, (A^{-1})^\dagger p \rangle \\ &= \langle (A^{-1})^\dagger p, v \rangle. \end{aligned}$$

Thus the covector $A \cdot \xi$ is represented by $(A^{-1})^\dagger p$. Since $A \in \text{SO}(3)$, we have $A^\dagger = A^{-1}$, and hence $(A^{-1})^\dagger = A$. Therefore, after this identification,

$$A \cdot (q, p) = (Aq, Ap).$$

It follows that, for $u \in \mathfrak{so}(3)$,

$$u_{T^*\mathbb{R}^3}(q, p) = \left. \frac{d}{dt} \right|_{t=0} (e^{tu}q, e^{tu}p) = (uq, up).$$

(2) We now identify u with a vector and compute Uq . Write

$$U = \begin{pmatrix} 0 & -c & b \\ c & 0 & -a \\ -b & a & 0 \end{pmatrix} \longleftrightarrow (a, b, c).$$

If $q = (q_1, q_2, q_3)$, then

$$\begin{aligned} Uq &= \begin{pmatrix} 0 & -c & b \\ c & 0 & -a \\ -b & a & 0 \end{pmatrix} \begin{pmatrix} q_1 \\ q_2 \\ q_3 \end{pmatrix} \\ &= \begin{pmatrix} -cq_2 + bq_3 \\ cq_1 - aq_3 \\ -bq_1 + aq_2 \end{pmatrix}. \end{aligned}$$

These are the coordinates of

$$(a, b, c) \times (q_1, q_2, q_3).$$

Thus $Uq = u \times q$. The same computation for p gives $Up = u \times p$. Therefore

$$u_{T^*\mathbb{R}^3}(q, p) = (u \times q, u \times p).$$

- (3) There is of course the tour-de-force path in which we do everything in coordinates and just confirm that the moment map and equivariance equations hold. But let's not do that.

I see two other ways: (1) try to use somehow that the symplectic form in $T^*\mathbb{R}^3$ is the differential of the tautological form, and (2) try to use that everything is linear.

Let's do (2). Recall that μ takes values in dual Lie algebra. Thus the vector $q \times p$ we are given is to be understood as a traceless antisymmetric matrix. The map $T^*\mathbb{R}^3 \ni (q, p) \xrightarrow{\hat{\mu}} (q \times p)u$ is most likely linear. So the exterior differential is... probably itself!

□

11. TORIC VARIETIES

For any compact connected symplectic manifold (M^{2n}, ω) with an action of a torus $\mathbb{T}^n$ and moment map $\mu : M \rightarrow \mathbb{R}^n = (\mathfrak{t}^n)^*$, the image of μ is a polytope. By [Guilleimin-Sternberg] and [Atiyah], this polytope is convex. Further, if the dimension of the torus is exactly half the dimension of the manifold, by [Delzant], the moment polytope determines the manifold, i.e. there is a one-to-one correspondence between Delzant polytopes and symplectic toric varieties. Moduli space of polygons is an example of this, see [KM95].

In fact, this situation of having a torus action of half the dimension is what we may call a “totally integrable system”.

12. (COMPLETELY) INTEGRABLE SYSTEMS

Let (M, ω) be a symplectic manifold. Recall that

- $\{f, g\} = \omega(X_g, X_f) = L_{X_f}g = -L_{X_g}f$.
- $X_{\{f, g\}} = [X_f, X_g]$

If $\{f, g\} = 0$ then the Hamiltonian flows of f and g commute.

Definition 12.1. Let (M^{2n}, ω) and $H \in C^\infty(M)$. (M, ω, H) is a *completely integrable system* (in Liouville's sense) if there exist functions $f_1, \dots, f_n \in C^\infty(M)$ such that

- (1) $H = f_1, \dots, f_n$ are linearly independent ($df_1, \dots, df_n$ are linearly independent in every point).
- (2) $\{f_i, f_j\} = 0$ for all i, j .

Note that $F = (f_1, \dots, f_n) : M \rightarrow \mathbb{R}^n$ is a submersion. For all $c \in \mathbb{R}^n$, $F^{-1}(c)$ are Lagrangian submanifolds. $F^{-1}(c)$ is invariant by the flow of X_{f_i} for $i = 1, \dots, n$.

Explanation: $F^{-1}(c)$ is a level-set submanifold of M , whose tangent space is generated by the df_i . These are all in the kernel of dF by the integrability condition, namely $dF = (df_1, \dots, df_n)$, $df_i(X_{f_j}) = \{f_i, f_j\} = 0$.

The idea is that finding these “conserved quantities” we manage to describe the Hamiltonian dynamics completely. Showing that a system is integrable is a victory. The moduli space of polygons is an example, and I think that so is $(\mathbb{C}^2)^{[n]}$.

There is another method of using “bihamiltonian actions” (two Hamiltonian actions yielding the same dynamics) /Lax pairs (defining the system using eigenvalues of matrices). Examples of this are Toda, Caolengo-Moser.

The following theorem says that the connected component of a symplectic manifold with an integrable system

Theorem 12.2 (Arnold-Liouville). *Let (M^{2n}, ω) , $H = f_1, \dots, f_n$ be completely integrable system and $F : M \rightarrow \mathbb{R}^n$, $F = (f_1, \dots, f_n)$. Let M_c be the connected component of $F^{-1}(c)$. Then*

- (1) *If M_c is compact, then $M_c \simeq \mathbb{T}^n = \mathbb{R}^n / 2\pi\mathbb{Z}^n = \{(\theta_1, \dots, \theta_n) \bmod 2\pi\}$.*
- (2) *In angular coordinates, the Hamiltonian flow is linear, that is, $\frac{d}{dt}^{-1}(t) = \alpha \in \mathbb{R}^n$ implies that $\theta(t) = \theta_0 + \alpha t$.*

The proof of item 1 is given by the following proposition putting $N = M$ and $X_1 = X_{f_1}, \dots, X_n = X_{f_n}$.

Proposition 12.3. *Let N be complete, connected, of dimension n . Let $X_1, \dots, X_n$ be linearly independent vector fields on N such that $[X_i, X_j] = 0$ for all j . Then $N \simeq \mathbb{T}^n$.*

Proof. First we present the steps:

- (1) Flows of $X_1, \dots, X_n$ define an action of $\mathbb{R}^n$ on N .
- (2) This action is transitive. Then $N \simeq \mathbb{R}^n / \Gamma$ where Γ is the stabilizer.
- (3) Since $\dim N = n$, then Γ is a lattice of $\mathbb{R}^n$.
- (4) Since N is compact, then Γ is of maximum rank, that is, $\Gamma = 2\pi\mathbb{Z}^n \subset \mathbb{R}^n$.

Let φ_t^i be the flow of X_i (which are complete since N is compact). Since $[X_i, X_j] = 0$, then $\varphi_t^i \varphi_s^j = \varphi_s^j \varphi_t^i$. Thus we have an action of $\mathbb{R}^n$ on N given by moving along the flows of the vector fields, that is

$$\underbrace{(t_1, \dots, t_n)}_{\mathbf{t}} \cdot x = \underbrace{\varphi_{t_1}^1 \circ \dots \circ \varphi_{t_n}^n}_{\varphi_{\mathbf{t}}(x)}(x)$$

To prove transitivity we can show that the orbit maps of this action are surjective. Indeed, the image of the orbit map is open since it is a local diffeomorphism (it's derivative is given by the vector fields, which are linearly independent!), and it is closed by a taking a point in the boundary and using the flow at this point (and commutativity of the flows).

To check that the stabilizer Γ of this action is trivial first note that all stabilizers are equal (not only conjugate) since $\mathbb{R}^n$ is abelian. Further, since the orbit map is a local diffeomorphism, Γ is discrete. Thus, it must be a lattice of rank $k \leq n$.

To show that in fact $k = n$ consider a linear map T that maps the generators of Γ to the first k canonical vectors of $\mathbb{R}^n$. By the following diagram, the arrow on the bottom is well defined as in fact a bijective local diffeomorphism, that is, a

diffeomorphism.

$$\begin{array}{ccc}
 \mathbb{R}^k \times \mathbb{R}^{n-k} & \simeq \mathbb{R}^T & \longrightarrow \mathbb{R}^n \\
 \downarrow p & \searrow \psi_x \circ T & \downarrow \psi_x = \text{orbit map} \\
 \mathbb{T}^k \times \mathbb{R}^{n-k} & \xrightarrow{\bar{T}} & N
 \end{array}$$

Thus the factor $\mathbb{R}^{n-k}$ must be a point since N is compact by hypothesis. $\square$

Item 2 from Theorem 12.2 is obtained by considering the flows on the latter diagram and is left as an exercise.

In the case that N is not compact we just conclude that it M_c is a product of a torus with an Euclidean space.

Perhaps the simplest example of an integrable system is given by taking an open ball about the origin $B \subseteq \mathbb{R}^n$ and $B \times \mathbb{T}^n = \{(I_1, \dots, I_n, \phi_1, \dots, \phi_n)\}$ with the symplectic form $\omega = \sum_{i=1}^n dI_i \wedge d\phi_i$ with $\{I_1, I_j\} = 0$.

The following theorem says that in fact the **symplectic form** of every level set of the kind M_c , which we just proved that is a torus, may is described locally as such a product.

Theorem 12.4. *There is a neighbourhood U of M_c and a symplectomorphism*

$$\begin{array}{ccc}
 \sum dI_i \wedge d\phi_i & \xleftarrow{\sim} B \times \mathbb{T}^n & \xrightarrow[r \simeq]{} U \subseteq M \xrightarrow{\sim} \omega \\
 \downarrow p & & \downarrow F \\
 B & \xrightarrow[\simeq]{} & F(U)
 \end{array}$$

The coordinates $(I_1, \dots, I_n, \phi_1, \dots, \phi_n)$ are called *action-angle* coordinates.

The case when $h(I)$ does not depend on the angle coordinates ϕ_i we can solve the Hamiltonian equations easily:

$$\begin{cases} \dot{I}_i = 0 \\ \dot{\phi}_i = -\frac{\partial h}{\partial I}(I) \end{cases}$$

so that $\phi_i(t) = \phi_0 + \alpha t$ for some constants α . This is why integrable systems are actually easy to solve!

13. VERTEX ALGEBRAS

Mônadas, álgebras sobre mônadas. Exemplo: ultrafiltros (Manes, 1976). Campo quântico, como serie de Fourier. Exemplo: Heisenberg algebra.

Problema: o producto de campos quânticos não é bem definido.

REFERENCES

- [KM95] Michael Kapovich and John Millson, *On the moduli space of polygons in the Euclidean plane*, Journal of Differential Geometry **42** (1995), no. 1, 133 – 164.